SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: MLA03

COURSE CODE: Reinforcement learning for Environment and Agent

COURSE OUTCOME COVERED

CO2: Apply Dynamic Programming methods to solve reinforcement learning

problems and derive optimal policies for decision-making tasks. (BL2, BL3)

Assessment Tool Weightage: 35%

Dr G. A. SENTHIL

QUESTIONS: 10 Total Marks: 50

Annexure A

Industry Problem Task Duration: 2.5hrs

Code of Conduct:

I T.GANESH(192425246) (Name / Reg No) certify that this submission is my original workand that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

T.GANESH

Signature of the student:

Question 1: Delivery Route Optimization in Logistics

1. Understanding of Industry Problem & Constraints

Logistics networks face dynamic routing challenges including real-time traffic jams, variable customer delivery windows, unpredictable weather conditions, vehicle capacity constraints, and fuel cost volatility. Traditional static routing models (e.g., Dijkstra, classic VRP heuristics) fail when operational conditions shift dynamically during transit.

2. Reinforcement Learning MDP & Engineering Design

We model the routing problem as a **Markov Decision Process (MDP)**:

- **State Space (S):** Current vehicle location (GPS coordinates), remaining package capacity, current time, time windows of unvisited nodes, live traffic density vectors, and driver working hours remaining.
- **Action Space (A):** Discrete choice of the next destination node to visit, or alternative transit speed/route adjustments.
- **Reward Function (R):**

$$R(s,a) = -(\alpha \cdot \text{Travel Time} + \beta \cdot \text{Fuel Cost} + \gamma \cdot \text{Late Arrival Penalty}) + \delta \cdot \text{Successful Delivery Bonus}$$

- **Policy (π):** A deep neural network policy $\pi_\theta(a|s)$ mapping current fleet state to the optimal next node assignment to maximize expected cumulative discounted return $G_t = \sum_{k=0}^{\infty} \gamma^k R_{t+k+1}$.

3. Solution Design & Continuous Learning

- **Modern Tools & Frameworks:** PyTorch / Ray RLLib using Deep Q-Networks (DQN) or Proximal Policy Optimization (PPO), integrated with Google Maps API / OpenStreetMap (OSM) for real-time environment simulation.
- **Continuous Learning:** As vehicles navigate routes, real-world experience tuples (s, a, r, s') are continuously collected into an Off-Policy Replay Buffer. The model retraining loop periodically updates the policy offline to adapt to seasonal traffic shifts, local road closures, and changing demand patterns, ensuring efficiency continually scales over time.

Question 2: Fintech Fraud Detection System

1. Industry Context & Fraud Mechanics

Financial institutions must detect adversarial transaction fraud in sub-millisecond latent windows without inducing excessive False Positives (which frustrate legitimate consumers and cause customer churn).

2. RL Formulation (Contextual / Multi-Armed Bandits & Deep RL)

- **State Space (S):** Transaction amount, user historical risk profile, device IP/fingerprint embeddings, geographic velocity (speed distance between consecutive swipes), and merchant Category Code (MCC).

- **Action Space (A):** a_0: Approve transaction; a_1: Flag for Multi-Factor Authentication (MFA); a_2: Decline transaction.
- **Reward Mechanism (R):**
  - **Approve Legitimate:** $+1$
  - **Approve Fraud:** $-$Transaction Amount $\times$ Chargeback Multiplier
  - **Block Fraud:** $+$Transaction Amount $\times$ 0.5
  - **Block/Flag Legitimate (False Positive):** -10 (Customer Friction Cost)

3. Exploration Strategies & Detection Accuracy

Fraud patterns evolve adaptively (adversarial shift). Using **Upper Confidence Bound (UCB)** or **Contextual ϵ-Greedy / Thompson Sampling**, the system systematically explores borderline transactions by routing them to step-up verification (MFA). This yields labeled ground-truth data without exposing high risk to the platform, continuously updating the model against emerging fraud tactics.

Question 3: Content Recommendation in Streaming Platforms

1. Industry Constraints

Streaming networks (e.g., Netflix, Spotify) face long-term user engagement optimization challenges where maximizing immediate clicks (clickbait) leads to subscription cancellations over time.

2. RL System Architecture

- **Delayed Rewards:** User satisfaction is not instantaneous. Immediate metrics (click/play) must be combined with delayed signals like total watch time percentage, rating completion, and 30-day retention/subscription renewal.
- **User Feedback & Non-Stationary Preferences:** User interest changes based on time of day, mood, and content fatigue.

3. Solution & Performance Analysis

- **RL Framework:** Actor-Critic Architecture (DDPG / SlateQ) where the Actor recommends a ranked slate of items, and the Critic evaluates long-term engagement potential.
- **Handling Delays & Drift:** Uses **Temporal Difference (TD) learning** with reward discount factors ($\gamma \approx 0.99$) combined with real-time vector embeddings updated continuously via online learning to adapt to evolving user taste profiles.

Question 4: Predictive Maintenance in Industrial Environments

1. RL vs. Traditional Approaches

Feature	Traditional Rule-Based / Thresholding	Reinforcement Learning Approach

Trigger Mechanism	Fixed sensor thresholds (e.g., Temp $> 90°C$)	Continuous state vector pattern recognition
Adaptability	Rigid; high rates of false alarms	Dynamic; learns multi-sensor degradation relationships
Action Optimization	Schedule maintenance at fixed intervals	Schedules dynamic maintenance to maximize remaining useful life (RUL)

2. Risk & Reliability Engineering Concerns

- **Exploration Hazards:** Random exploration in real manufacturing environments can cause catastrophic physical machinery breakdown.
- **Mitigation Strategy:** Safe RL via **Constrained Markov Decision Processes (CMDPs)** and off-policy training inside high-fidelity **Digital Twin simulators** (built on Ansys or NVIDIA Omniverse) before deployment.

Question 5: Smart Urban Transportation & Bus Scheduling

1. Problem Formulation

Public transit systems suffer from passenger congestion, "bus bunching" (uneven headway intervals), and inefficient fleet utilization due to unpredictable traffic bottlenecks.

2. MDP Architecture

- **State Space (S):** Current positions/speeds of all active buses, passenger count at each bus stop (via automated passenger counter sensors), road congestion indexes, and weather status.

- **Action Space (A):** Hold bus at stop for X seconds, speed up/slow down travel speed target, reroute to express bypass, or inject backup fleet.
- **Reward Function (R):**

$$R = -\left(\sum \text{Passenger Waiting Time} + \sum \text{Passenger Transit Time}\right) - \text{Fuel Consumed}$$

3. Real-Time Data Integration

By streaming real-time IoT GPS feeds and automated passenger counts via Kafka pipelines, Deep Q-Networks (DQN) continuously dynamically adjust bus dispatch intervals, eliminating bus bunching and minimizing overall commuter delay.

Question 6: Dynamic Pricing in E-Commerce

1. Market Mechanics

E-commerce dynamic pricing requires balancing immediate profit margins against long-term sales volume, competitor price matching, and consumer price elasticity.

2. RL Formulation

- **State Space (S):** Current inventory stock level, days until holiday/season end, competitor pricing, conversion rate trends over past 24 hours, and page view traffic volume.

- **Action Space (A):** Discrete price adjustment steps (e.g., $-10\%, -5\%, 0\%, +5\%, +10\%$) or continuous pricing bounds.
- **Reward Function (R):** Total Gross Profit = $(\text{Selling Price} - \text{Cost}) \times \text{Units Sold}$.

3. Exploration vs. Exploitation Balance

- **Challenge:** Over-exploring price increases causes immediate loss of customer trust; over-exploiting lower prices erodes margins.
- **Engineering Solution:** Apply **Contextual Thompson Sampling** or **Entropy-Regularized PPO (Soft Actor-Critic)** to maintain controlled exploration within defined safe price bounds set by business rules.

Question 7: Smart City Waste Management Systems

1. MDP Components & Optimization Focus

Urban waste management faces inefficient fixed collection schedules leading to empty bin collection or overfilled, unhygienic bins.

```
+------------------------------------------------------------+
|                    SMART CITY MDP                          |
|                                                  |
|   [State Space]  --->  [RL Agent (DQN/PPO)]  --->  [Action
Space] |
|   - Fill Level %       Dynamic Routing            - Collect Bin|
|   - Bin Location        & Dispatch Policy         - Skip Bin   |
|   - Traffic Context                          - Alert Ops  |
```

| |

| [Reward Function] = - (Travel Distance + Fuel) + Cleanliness |

+--+

- **States (S):** Fill level percentage of smart bins (via IoT ultrasonic sensors), days since last collection, localized traffic condition, dynamic weather (rain increases odor/health hazard).
- **Actions (A):** Assign collection route to truck k, skip bin, or trigger emergency alert for instant collection.
- **Reward (R):** Penalty for fuel/route length plus high penalty for overfilled bins ($> 90\%$), balanced by reward for maximizing average cleanliness score per route km.

2. Continuous Updates & Efficiency Gains

Continuous ingestion of telemetry streams allows reinforcement algorithms to dynamically update route maps daily, optimizing fuel consumption by up to 30% compared to static routes.

Question 8: Network Bandwidth Allocation in Telecom

1. Telecommunications Challenge

5G/6G dynamic network slicing requires real-time bandwidth allocation under fluctuating, unpredictable traffic spikes (e.g., stadium events, streaming surges) while maintaining strict Quality of Service (QoS) Service Level Agreements (SLAs).

2. RL Architecture

- **State Space (S):** Latency across slices, packet drop rate, current user connection density, buffer queue lengths, and spectral efficiency.
- **Action Space (A):** Reallocate X MHz of bandwidth from Low-Priority (IoT/Browsing) to High-Priority (Emergency/Autonomous Driving/HD Video) slices.
- **Reward Function (R):**

$$R = \sum \text{Throughput} - \lambda_1(\text{SLA Violations}) - \lambda_2(\text{Bandwidth Reallocation Cost})$$

3. Dynamic Adaptation

Using **Model-Free Deep Deterministic Policy Gradient (DDPG)**, the telecom agent dynamically re-allocates sub-band channels in millisecond timeframes, adapting instantaneously to unpredictable traffic surges without human operator intervention.

Question 9: Portfolio Management in Finance

1. Advantage over Traditional Portfolio Models (Markowitz MPT)

Modern Portfolio Theory assumes stationary returns and normal distributions. RL dynamically adapts asset allocations to non-stationary market regimes, tail-risk events, and structural economic shifts without rigid statistical assumptions.

2. System Architecture & Constraints

- **State Space (S):** Asset price momentum indicators (RSI, MACD), asset correlation matrices, volatility indexes

(VIX), order book depth, and macroeconomic sentiment embeddings.

- **Action Space (A):** Continuous portfolio weights vector $W = [w_1, w_2, \dots, w_n]$ such that $\sum w_i = 1$.
- **Reward Function (R):** Dynamic Sharpe / Sortino Ratio penalized by transaction slippage and trade execution costs:

$$R_t = \frac{\mathbb{E}[R_p - R_f]}{\sigma_p} - \text{Transaction Fees}$$

3. Risk Mitigation

To avoid dangerous drawdowns during high volatility, policies are trained using **Soft Actor-Critic (SAC)** with conditional Value-at-Risk (cVaR) constraints embedded directly in the reward function.

Question 10: Renewable Energy Grid Management

1. Justification for RL in Renewable Systems

Solar and wind power generation are inherently stochastic and intermittent. Combining fluctuating supply with dynamic grid demand requires an adaptive controller capable of real-time balancing between generation, battery storage, and grid loads.

2. RL Framework Definition

- **State Space (S):** Current solar/wind generation, battery State of Charge (SoC), real-time energy spot price, forecasted 24-hour weather, and microgrid consumer demand load.

- **Action Space (A):** Charge battery, discharge battery to grid, or curtail excess renewable generation.
- **Reward Function (R):**

$$R = \text{Revenue from Grid Sales} - \text{Energy Procurement Cost} - \text{Battery Degradation Penalty}$$

3. Handling Environmental Uncertainty

Through training over decades of synthetic historical weather profiles, the RL agent develops robust strategies—learning to pre-charge storage ahead of expected low-generation peak-demand windows, minimizing blackouts and optimizing economic returns despite severe environmental unpredictability.